The Law of Graphic Equalisation — Mind and Self Grounding

Frames, the Run and the Record, Free Will, Self-Models, Personal Identity, and Death

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Abstract

This companion to Graphic Equalisation develops the mind-and-self groundings of the change-substrate. Its central move is that a frame is real graph data that *runs*, and that everything here follows from the run and its record. A rock and a universe are both frames, so anything a rock does without is not compulsory for a frame. Recording spatialises a succession into a static held axis, giving the run/record axis whose poles are Conway (pure running, computationally irreducible) and a movie (pure record), with real frames the machinima between; the past can be a record but the future is uncomputed, so it is open as *unrun*, not as indeterministic. Free will is then the in-frame run — a fully recorded, zero-randomness story demonstrates it, its characters free in frame while the reader foreknows — so freedom needs no indeterminacy. Imagination is real: there is no non-actual frame, and the only difference between the imagined and the perceived is frame-relative grounding. Self-models are incomplete by construction: a frame carries its own trajectory rather than reading it, so to treat that course as an object at all it must build a model of itself, under a finite aperture. The incompleteness is structural rather than pathological, and it divides by kind — a limit of *expression* leaves residual and is inferable from trace, while a limit of *aperture* leaves nothing and is inferable only by metacognition — so the operative distinction is calibrated versus uncalibrated, not complete versus incomplete. Death is the self's rules ceasing to be re-runnable in the same coherent self; personal identity is internal coherence (with an instance a supergraph node), which places gradual replacement, teleporter, backup, and fission on one axis; and death is relative to a chosen coherence-boundary, with many real deaths and none absolute. The claims throughout are information-level and structural.

1 Scope

The method is to treat philosophy of mind as applied physics: once a frame is graph data that runs, free will, identity, and death are read off the run and its record, not argued as separate doctrines. This paper is a grounding, not a psychology and not a metaphysics asserted from within. It inherits Graphic Equalisation's position that the claims are about information — distinction, frame, admissibility, realisation, and now the *run* of a frame and its *record* — of which mind is one embodiment. The strong claims are structural: that free will is the in-frame run and needs no indeterminacy; that imagination is real and differs from perception only in

grounding; that every self-model is incomplete by construction, so the operative distinction is calibrated versus uncalibrated rather than complete versus incomplete; that personal identity is internal coherence (Graphic Equalisation's SELF identity), distinct from the supergraph node that instances it (its SUPERGRAPH identity); and that death is the loss of re-runnability in the same coherent self, relative to a chosen coherence-boundary. The weak part is anything operational: no decision procedure for a given choice, no test for when re-runnability or coherence has in fact been lost, no threshold for how gradual a change stays "the same" self, and no account of the subjective, felt, or phenomenal character of any of it — consciousness and qualia are out of scope. Each grounding is a *promoted* constraint exposed to falsification, in the manner of Graphic Equalisation's collapse bridge, not a result asserted from within.

2 The minimal universe

A *rock*, in the sense used throughout, is an inert frame: a graph whose rules are present but not firing, so it has no succession and no time of its own. Things happen to it from its supergraph, but in its own frame it merely is (the physics companion develops this as why a rock has no time). Everything here is a graph, so a rock and a universe are one kind of thing, differing in scale, not in category (Graphic Equalisation's self-similarity: a primitive is graph-that-runs at every scale). A rock *is* a universe — the minimal one — and that makes it an existence proof about universes in general.

Proposition 2.1 (What the rock shows is not compulsory)**.** *Anything not compulsory for a rock is not compulsory for a frame, and the universe is one frame among others. The rock does without time (it is inert, no succession), without cycles or closure, without inference or observation, without history or records, without self-originated change, and without a running arrow — so none of these is compulsory for a universe.*

The inference is one-directional, and only that direction is claimed. A single inert example bounds the compulsory set from *above* — it shows what can be dispensed with — and cannot establish that what the rock happens to retain is necessary. So the rock is a non-necessity argument, not a derivation of a core.

What is compulsory is inherited from Graphic Equalisation, which keeps almost nothing: a container relation across which rules run. The rock is consistent with that and does not weaken it, because a rock's rules are inert *in its own frame* while its container's are not — what happens to the rock (weathering, being moved) is its supergraph's rules running across exactly that container relation. An inert frame is therefore one whose own rules do not fire, not a counterexample to rules running.

Nor does that demotion cost the argument anything, because *frame* is the category throughout. The universe is a frame and a rock is a frame; they differ in scale, not in kind, which is what self-similarity asserts. So the inference never needed "universe" as a separate qualification — it runs over frames, and reads: anything a frame can be without is not compulsory for a frame. The rock's inertness is not a defect in the example but the whole of its content.

This is the general form of "time is optional," and it licenses the contingent-not-necessary verdicts of the physics companion. Wherever a shape there was called contingent rule-content rather than necessity — the number of spatial dimensions, the gauge group, the dimensionless constants, the arrow, the value of c — the rock is the reason: the minimal universe does without it, so it cannot be compulsory. Physics ordinarily takes time, causality, and the second law's arrow as features a universe must have; the rock shows they are features an *active* universe has.

3 Running and recording: Conway, a movie, and the machinima between

Recording spatialises time. A succession held as a record — an MRI of a rock, a film strip, a worldline drawn on paper — lays the temporal order out as a static structure, a +-like axis (the block's extra dimension). This reads the metric signature: the − axis is the one being *lived*, the frame's own succession run by its cycles, open at the frontier; the + axes are *held* — spatial positions, and any recorded time. The sign tracks lived-versus-held, not a property intrinsic to the axis; the block feels dead because a recorded succession is not run.

The extremes of the run-versus-record axis are Conway and a movie. Conway's Game of Life is pure running: a rule and a seed, everything computed live, nothing recorded, and no way to know the outcome but to run it (computational irreducibility). A movie is pure record: every frame laid out, played back, nothing computed — the block universe, known by reading rather than running. Between them is scripted machinima — a running engine following a predetermined script, part computed live, part laid down in advance — and that is where real frames sit: rules fire (Conway-like) while history is held (movie-like). It is the same axis as generation-versus-record: store the generator and run it (Conway, small but irreducible) or store the frames and read them (a movie, large but immediate), with the world's own compressibility deciding which is cheaper.

Proposition 3.1 (The past can be a movie; the future is Conway)**.** *A block is only ever of the past. What has been run can be recorded and laid out — a movie of the succession so far, readable without re-running. What has not been run cannot be recorded, because there is nothing yet to record: the future is determinable by the rules but* uncomputed*, Conway-style, so it is not laid out and cannot be read ahead — you must run it. The open frontier of Axiom* −1 *is therefore not indeterminism but unrun-ness: the future is open because it is uncomputed, not because nothing fixes it, and you cannot record what has not run.*

4 Free will and the three frames

Proposition 4.1 (Free will is the in-frame run; a story demonstrates it)**.** *Free will is a frame running its own decision. In a frame's own run the decision is live and unrun-to-it — the frame must run its deliberation to reach the outcome — and that running* is *the choosing: the outcome is fixed by the rules yet knowable only by running, so the frame experiences computing it as choosing it. This needs no indeterminacy, and a story is the perfect demonstration. A book or a movie is a fully recorded succession — every word or frame fixed, zero randomness, readable ahead by the reader — and yet its characters have complete free will: they deliberate, choose, and are responsible, in frame. Both hold at once because they are two frames. The character's freedom is its in-frame run; the reader's foreknowledge is a superframe holding the finished record (a movie), and holding the record never reaches into the character's run to remove the freedom. So foreknowledge is not unfreedom — not even complete foreknowledge, the reader who knows the ending or the author who wrote it — because knowing the recorded run and being the live run are different frames. Randomness would only subtract: a random run is not more free, it is less the character's own. And responsibility survives for the reason the freedom does — the run is the character's own (self-change is the only change; the rules running are the character) — so a determined, recorded, even foreknown act is still that character's act.*

Remark 4.2 (Author, character, reader: unrun-ness is absolute). A story stacks three frames on one succession. The author *runs* it — writing is Conway, generated live, and the author must write to know the ending (authorship is computationally irreducible; the commonplace that "the characters took on lives of their own" is the author unable to read ahead their own creation). The character *lives* it — the in-frame run of Proposition 4.1. The reader *replays* it —

the finished record, a movie. One succession, three frame-readings: the run/record axis is a fact about frame, not about the system. And the author settles the objection to Proposition 3.1 that a high enough superframe might read the future ahead — even the creator gets no shortcut. The author is Conway to their own book while writing it; the movie exists only *after* the run, for everyone including the author. Foreknowledge is always post-run record-holding, never pre-run read-ahead. For an irreducible rule unrun-ness is absolute: no frame, creator included, reads an unrun future without running it; every frame either runs it or replays an already-run record.

Remark 4.3 (The reader runs and holds at once, and this is standard). The reader looks like a counterexample: the account needs the character's frame genuinely to run, and needs the reader to hold a finished record — and if the character's frame runs *in* the reader, the two frames appear to collapse into one. They do not, and the point does not need arguing here, because three separate literatures settled it independently.

The cleanest statement of the running half is Currie's: a reader of fiction simulates the state of a hypothetical *reader of fact* (Currie, 1997; Currie and Ravenscroft, 2002). What runs in the reader is therefore a simulation, and what it simulates is someone taking the events as real — so running the character's frame is not in competition with holding a text, because the running is itself an offline state rather than a belief. Currie's figure does not supply the *foreknowledge*, though: his reader-of-fact meets events as news, in order, and knows no more of the ending than any reader of a report. That half comes from elsewhere, and rhetorical narratology supplies it by dividing the reader rather than the frame: Rabinowitz (1977) distinguishes the *narrative* audience, for whom the narrated events are real, from the *authorial* audience, which knows the text is made, and these are positions one actual reader occupies simultaneously (Polvinen and Sklar, 2019). The two frames were never reader-versus-character. And the cognitive architecture supplies the mechanism: pretence representations are processed by the same inference machinery that operates over beliefs — a single code (Nichols and Stich, 2000) — while *quarantining* confines their effects to a circumscribed domain (Gendler, 2003). Same machinery is not same frame. Walton (1978) states the resulting two-level structure exactly: what runs in the appreciator is real (his "quasi-fear" — the tensed muscles, the quickened pulse) and it is that real state which makes it make-believe that he is afraid, with full knowledge that the slime is not. Literary theory has a name for the compound, *aesthetic illusion*: immersion counterbalanced by a latent rational distance, the awareness of the difference between representation and reality (Wolf, 2014).

Nor is it only characters that are modelled. A reader builds a model of the *author* as well, and that construct is long established: Booth's *implied author*, the version of the writer a reader inevitably assembles from the text (Booth, 1983), which the standard treatment takes to have both an objective side, grounded in textual indexes, and a subjective one, since those indexes are weighed differently by each reader (Schmid, 2014). Genette (1988) puts it most economically — the implied author is everything the text lets us know about the author — though the sentence is an opponent's maximum concession rather than an endorsement, appearing on the same page as his warning not to turn the idea of the author into a narrative agent, in a work concluding the concept is not narratology's business. It is quoted for its precision, not for its support. Two further constructs name the same object from neighbouring positions and should be recorded with it: the *represented author*, defined as the reader's representation of the author of the text, in a programme built expressly for the empirical study of literary response (Bortolussi and Dixon, 2003), and the *postulated author*, offered as a regulative ideal for interpretation rather than as a person (Nehamas, 1981). Neither is generative, so neither bears on the composition claimed below; both bear on how routine the construct is. Hypothetical intentionalism makes the reasoning explicit, locating a work's meaning in what an appropriate audience would most reasonably hypothesise the situated author to have meant by composing this text (Levinson, 2002). That viewers reach for an author at all has some empirical support, though only one half of it survives the argument below: when people distinguish professional abstract works from superficially similar ones by children and animals, they justify the professional choices by

reference to the *artist's intentions* (Hawley-Dolan and Winner, 2011). The discrimination itself shows nothing here, for the reason given shortly; the appeal to intentions is what indicates a model of a maker.

The mode is the one this paper has already named. A frame cannot read its author's frame; it has only what that frame produced, and so infers it — the +1 direction of the level stack, run on a person rather than a container. Sustained output makes that inference rich, since a generator sampled at length discloses a great deal of itself whether or not it means to.

That such a model is *generative* — runnable forward on inputs the author never addressed — is the part that had to be checked hardest, and the check did not confirm the claim but reduced it twice, the second time on reading the source rather than a summary of it. Narratology supplies generativity in full, and in the same theorist quoted above.

Genette (1982, 82–83) states as a principle that *it is impossible to imitate a text directly.* Imitation must pass through something abstracted from the corpus, which he names in his own terms: "a *matrix of imitation* is inevitably interposed, which is the *model of competence* or, if one prefers, the *idiolect* of the imitated corpus." The framing is explicitly the generative grammarian's — competence against performance, *langue* against *parole*, an idiolect being "not a message but rather a language: a code wherein the specificities of the message have been made fit for generalization" — and the operative claim is that "the essence of competence is the ability to generate an *unlimited number of correct performances*." His worked example runs the model forward: from scattered Balzac locutions he abstracts the formula *x, that y of z* and then produces on it a phrase Balzac never wrote. So the extraction of a generator from a corpus, and its use on material the source never treated, is set out with a mechanism and an instance, in 1982.

What survives is narrower still, and turns on what *correct* means in that sentence. A pastiche's correctness is stylistic well-formedness: Proust's new Balzaceme is a good one because it is a locution Balzac could have written, and no question arises of whether Balzac would have held the thought. The claim made here has a different success condition — what a particular person would have made of something they never addressed, right or wrong in ways that can be checked against them — and that is why it needs a model of a *person* rather than of a manner, and why the empirical case discussed below scores its answers against the living philosopher's own. Genette's standing objection to personifying the implied author is consistent with all of this and is not an obstacle to it: he declined to take the step, having built everything the step requires.

So the generativity is prior art, fully; only the personification and the truth condition attaching to it are not. That is the reverse of how an earlier version of this passage divided them, and the status is worth stating exactly, because none of it is a new posit. It is a composition of three things each established independently. An author-model is a model. Models can approach the behaviour they model arbitrarily closely, given enough of it to work from. And analysis by synthesis runs in both directions: a recogniser built that way recognises *by* generating candidates and comparing them, so it contains a generator as a matter of construction rather than as an addition to it. One further step is needed and should be marked rather than glossed: the generator inside such a recogniser is *conditional*, driven by the item being recognised and closed inside a comparison loop, whereas running forward means conditioning it on something no percept is supplying and accepting its output without a target to check against. That the same generator can be driven by novel conditioning is the paper's own premise, not one inherited from the perception literature. That architecture is not speculative. It is the modern form of the predictive account already drawn on above — perception as the inversion of a generative model, predicted input compared against actual (Friston, 2010) — and it has been tested directly on people: three-dimensional shape perception is best explained as analysis by synthesis, matching human observers in accuracy and response time better than the alternatives, and alone correlating with human performance on the hard discriminations (Yildirim et al., 2024). A

generative author-model follows from those, and needs no further machinery.

What that argument does not establish is that any particular reader's model is of this kind, and the distinction matters because the obvious evidence does not decide it. Recognising a style requires no model of its author at all: pigeons trained on Monet and Picasso discriminate the two, generalise to unseen paintings, and generalise across artists — Monet to Cézanne and Renoir, Picasso to Braque and Matisse (Watanabe et al., 1995). Discriminative recognition is available without any generator, so neither recognising a voice nor sensing that a pastiche is wrong shows anything about which kind of model is in play. The empirical position is in fact weaker than that, and in the direction that helps this argument rather than hurting it: asked to tell poems by well-known poets from machine imitations of them, non-expert readers scored *below* chance and rated the machine's poems as more human than the originals (Porter and Machery, 2024). Ordinary readers do not reliably manage even the discriminative task, so nothing can be inferred from it in either direction. What would is the forward direction — what the author would have made of something they never saw, right or wrong in ways others can check — and the honest report is that it has not been measured. A sweep of the indexes returns machine stylometry and nothing on human forward-prediction of an author, so the claim here is *untested* rather than supported or contradicted, and an earlier version of this sentence described a distribution across readers that no one has established. The nearest evidence is one step to the side and concerns characters rather than authors: readers who learn a protagonist from text can be shown to imagine that protagonist in situations the text never described, which those authors themselves describe as a personality model used to predict behaviour in novel situations (Hassabis et al., 2014) — from constructed two-trait protagonists, and decoded neurally rather than scored for accuracy. Readers' own accounts point the same way, being framed by the researchers who collected them as an expectation of what a character would say in a given situation (Alderson-Day et al., 2017).

The honest summary is therefore not that the generative reading is unsupported, but that its *components* are supported and its *composition* is this paper's. One influential line in narratology holds that the implied author does not communicate and must not be personified (Genette, 1988; Kindt and Müller, 2006), which is a claim about an interpretive construct's role in a theory of narrative; it is not an argument that a model of a generator cannot be run, and the two are easily conflated. It should not be called the mainstream, and an earlier version of this passage did: the question is under live dispute rather than settled — an entire journal issue was given to it (*Style*, 2011) — and Phelan, Shen, Chatman and Rabinowitz (2011) argue the other way. The last of those matters here specifically, because this section has already drawn on Rabinowitz approvingly, and attributing to a consensus a position he argued against in print would have been citing him against himself.

It has also been attempted. Schwitzgebel et al. (2024) fine-tuned a language model on one philosopher's complete works, put ten questions to the model and to the living philosopher, and had experts on his work try to pick out his answers blind from five candidates. They identified them about half the time against a 20% baseline — well above chance, and well below the 80% the authors had predicted — and on two questions more of them picked a machine answer as the author's own than picked his. The design is the forward one: the questions were fresh and the author himself supplied the ground truth. It does not cleanly separate the two capacities, though, and the authors say so: expert performance was highest on two questions carrying what the authors call a superficial tell — the philosopher wrote first names where the model wrote surnames — which is discrimination doing the work rather than the model's. Related work builds runnable models of specific people, from their writing (Wang et al., 2025a) and from biographical experience (Shao et al., 2023) and simulates individuals well enough to transfer to tasks no training data covered (Park et al., 2024). What none of this literature does is connect such models to what a *reader* does; it builds author-simulators as engineering, without reference to the implied author or to any claim about ordinary comprehension. The bridge is what is

claimed here, not the artefact.

How good such models get is a separate and largely empirical question, on which the same literature reports a split worth noting: what transfers most readily is topical content, while idiosyncratic voice and argumentative shape are harder, and the characteristic failure has been named and measured as *caricature* — the authors' own term, defined by failed individuation together with exaggeration of whatever is distinctive (Cheng et al., 2023), though measured there on demographic personas rather than on individuals. Where fidelity to a particular individual has improved, it has come from richer data about that individual rather than from larger models, which is what the leaked-sampling reading of Section 8 would expect. Nothing here turns on the state of any current system.

Two qualifications, both of which the substrate should expect. The engagement is not uniform across people: measured individually, mentalizing and motor engagement during narrative comprehension *trade off* against one another rather than rising together (Nijhof and Willems, 2015), so how one runs a character is itself a position on several axes rather than one capacity present or absent. Two limits belong with that, since it is the only measurement carrying this claim: the correlation is reliable in one region rather than across the network, and the eighteen participants *listened* to the excerpts, the authors having chosen listening over reading, so it is evidence about narrative comprehension and only indirectly about readers. And the separation is functional rather than metaphysical, and defeasible: quarantine leaks (Gendler, 2006), and readers report characters' voices persisting into and commenting on their own lives afterwards (Alderson-Day et al., 2017) — reported by about a fifth of those who wrote freely about their reading, and described by the authors as relatively rare across the full sample, with under four per cent reporting it most or all of the time. The phenomenon is real and is not the ordinary case, which is what the argument needs: quarantine that leaks sometimes is still quarantine. The claim is therefore that the frames are functionally separated, which is what the argument needs, and not that they are hermetically sealed, which is false.

Remark 4.4 (Cognition under the same treatment). Cognition is compressed by any measure — a brain takes on the order of a fifth of a body's resting energy for a fiftieth of its mass, and an energy budget for signalling has been computed in detail (Attwell and Laughlin, 2001) — for rodent grey matter, with the whole-body figures quoted there from earlier work rather than established by it. The stronger claim, that neural arrangements are forced by that budget rather than chosen, is Sterling and Laughlin (2015)'s thesis and not the budget paper's. Two constraints can then be named in the way Graphic Equalisation's convergence result requires, with the forcing constraint stated rather than gestured at, and only two; neither is supplied by either citation, and both are ordinary neuroscience. Conduction takes time over distance, so there is no central synchroniser and coordination is local and coupled — the same concession distributed engineering made, for the same reason. And bandwidth is scarce, so what is not predicted is what gets sent.

Two further candidates are excluded. That cost is inherited along provision is this account's own thesis (the substrate paper's recursion result), so offering it as an independent constraint on brains would be circular; and that introspection does not reach the machinery is not a constraint but the conclusion relabelled.

What this licenses is narrower than *cognition is the same*. Two structures under the same binding constraints land on the same arrangements over the region those constraints touch, and no further: cognition is not thereby computation, all cognitions are not thereby alike, and nothing here says a brain is a computer.

Remark 4.5 (A popular understanding equalises, and that is why it drifts). The evolutionary claims above read very differently depending on who is reading them, and the difference is not one of expertise so much as of aperture — which makes it an instance of the same machinery rather than an aside about communication.

The account of evolution most people hold is close to Darwin's: variation, competition, sur-

vival of the better fitted, over long stretches. Working biology in 2026 is not that. Developmental bias, phenotypic plasticity, inheritance beyond the gene, horizontal transfer, symbiosis and constructive neutral processes are all ordinary parts of the working picture, and the argument the previous remark takes a side in is an argument *within* that picture rather than an attack on it. So *natural selection is half of evolution* is heterodox against one aperture and unremarkable against the other, and a reader's reaction to it says which one they are holding.

Why the gap persists is the interesting part, and it is not that the explaining has been done badly. A popular understanding is a structure under compression like any other, and Darwin-level evolution is *well-formed for the purposes most people put it to*: it accounts for enough, it is cheap to hold, and almost nothing anyone meets in a day produces a residual it cannot absorb. It has equalised, and it persists for exactly the reason the horseshoe crab's form does — nothing is forcing a write. The working picture is the overshooting case, kept changing by a residual the field manufactures deliberately.

That predicts something checkable and slightly counterintuitive: the gap between popular and expert understanding of a field should track *daily exposure to residual* rather than education or communication effort. Where a lay frame is corrected constantly by contact — weather forecasting, where people check a prediction and see it fail — the gap is small and the lay frame keeps up. Where it is never contacted in a way that fails, the gap widens no matter how much explaining is done, because explanation supplies no residual: the frame is not wrong for what it is used for, so nothing in it has to give.

Two ends. This is not a claim that the popular account is ignorance — it is well-formed for its work, which is what makes it durable, and calling it deficient would be measuring it against work it is not doing. And expert consensus is not thereby correct; it is a frame under different pressure, which is a fact about its residual and not a warrant.

For this paper the practical consequence is small and worth stating: the evolutionary material is addressed to the working aperture, and a reader who finds it radical is invited to check it against the field rather than against what they were taught.

5 Imagination, grounding, and reality

Proposition 5.1 (There is no non-actual frame; imagination is real)**.** *Imagination is not the running of a non-actual frame — there is no such thing. Every frame that runs is real graph data really running, so actuality is not a property some frames have and others lack; every frame is actual. An imagined dragon, a planned move, a character in a book is a real frame, really run (real computation), not a pretend one. The only difference between an imagined frame and a perceived one is frame-relative grounding: what the frame is anchored to, relative to another frame. A perception is grounded in the observation frame (sensory-anchored); an imagination is grounded internally, in generation or in a held or communicated record. Both are real; they differ in what they are grounded in, not in whether they are. So "is it real?" is malformed as a question about actuality — everything that runs is — and the well-formed question is "what is it grounded in?": observation, memory, generation, or a communicated record. A shared fiction is a real frame grounded in a communicated record (the book of Remark 4.2); a plan is a real frame grounded in projection. This is why imagination does work — planning, design, and counterfactual reasoning are real frames really run, not pretence — and why fiction moves: the character's frame runs, really, in the reader. Death, in the same terms, is not a frame going non-actual (nothing does) but a run that stops, leaving frames grounded only in a record and re-runnable by others; reading and remembering are that re-running — a record-grounded frame run again into a live grounding.*

Predictive-processing accounts reach a closely related conclusion by another route. Clark (2016) argues that perception, dreaming, memory and imagination are expressions of a single

mechanistic ploy, with imagery arising when the same generative machinery runs top-down, insulated from the driving sensory signal — one kind of running, differing in what anchors it, rather than a second and weaker faculty producing contents of a lesser kind. Seth (2019) makes the perceptual half explicit, treating perception as predictive construction rather than transcription of input (the "controlled hallucination" formulation, which he credits to earlier sources rather than claims as his own). The routes are not the same: those are empirical claims about how one substrate is built, while the claim here is the structural one that grounding rather than actuality is what the imagined/perceived distinction consists in, and it would stand for a substrate built on different machinery. What the convergence supplies is the load-bearing part — that the difference is not a property carried in the frame or the signal.

Remark 5.2 (What the difference is, and how a substrate detects it). Grounding is what the difference between an imagined and a perceived frame *is*. How any particular substrate *detects* that difference is a separate question, answered by that substrate's implementation, and the two must not be conflated. Human reality-monitoring is the instructive case: imagined and perceived signals are carried in a common representation, and the reality judgement is made by a threshold on subjective signal *strength* rather than by reading provenance off the signal (Dijkstra et al., 2022; Dijkstra and Fleming, 2023). That is a fact about one implementation, not a correction to the account — and it is what the account predicts. A frame does not have its own grounding directly available to it, grounding being a relation to what lies outside the frame rather than a property carried within it, so any substrate that must sort its own frames has to use a proxy; signal strength is the proxy human brains use. The characteristic failure follows immediately, since a proxy dissociates from what it proxies for: misclassification is predicted exactly where strength matches and grounding differs, which is what is measured. It also fixes how the tagging language elsewhere should be read: a frame that sorts its own frames by a proxy holds no provenance label, so a "grounding tag" is an abstraction over whatever proxy does the sorting rather than a mark carried on the frame, and a mistagging is that proxy dissociating. A substrate using a different proxy would fail differently while the underlying distinction stayed the same. Note also that this is a first-order misclassification and not the second-order failure of Section 6: a frame can be perfectly aware that its proxy may mislead it and still be misled.

Proposition 5.3 (Grounding has no bedrock; observation is supergraph-contact)**.** *Grounding is frame-relative all the way down: a frame grounded in a frame grounded in a frame, with no absolute, self-grounding frame beneath them all. There is no bedrock "actual" layer — the ground is the coherent running structure, frame to frame, not a base. Yet one grounding is privileged, relationally and not absolutely:* observation *is a frame's contact with its supergraph, the boundary where the container deforms it with changes it does not author (Graphic Equalisation's dissolved pointer — you change yourself, the supergraph deforms you — with perception the deformation side). Observation is special because it is* yours*, your container-boundary, not because it is "the real." It follows that reality-testing is not actuality-checking — there is no non-actual frame to catch (Proposition 5.1) — but* supergraph-pushback*: does the container deform you as if the frame were grounded there? You walk into the chair (pushback: observation-grounded) and not into the dragon (no pushback: imagination-grounded), both real frames, only one with the container pushing back. Sanity is then a grounding-tag problem, not a reality one: keep observation-grounded frames tagged as observation-grounded and imagination-grounded ones as imagination-grounded. A hallucination or delusion is a grounding-tag error — a real, internally-grounded frame mistagged as observation-grounded, so the self expects a pushback that will not come — not a belief in something non-actual, since nothing is. The position is thus anti-foundationalist (no absolute ground) without being flatly coherentist (observation is privileged, as your supergraph-contact).*

That combination — no bedrock, and no loss of realism for lacking one — is not peculiar to this substrate: the standard worry that dropping a foundation takes realism with it is already

declined by perspectival realism, an established position in philosophy of science arrived at independently of anything here (Massimi, 2022).

6 Self-models: incomplete by construction, calibrated by choice

A frame does not read its own trajectory. It *carries* it — the standing structure is what the frame is, not something it holds at a distance — so to treat its own course as an object at all, a frame must build a model of itself. The incompleteness is not merely a consequence of the aperture being finite — a finite frame could in principle be modelled exhaustively — but of the model being a proper part of what it models. Modelling the frame includes modelling the model, and the apparatus doing the modelling has strictly less capacity than the whole it belongs to, so the fit cannot close. Three features of that model follow from the substrate rather than from any fact about minds.

It is *incomplete*, because it is built under a finite aperture, and what falls outside is NULL rather than zero: an absent distinction does not present itself as a blank to be filled, it does not present at all. It is *revisable only at a price*, because altering standing structure is the paid operation while retaining it is free (Narrative momentum), so coherence-preserving readings are cheap and re-writes are not. And it is *indirect about its own subject*, since a frame is readable from outside in a way it is not from inside — which is why another frame often models one's trajectory more accurately than one does oneself, without any appeal to self-deception — a consequence of access mode, though not an entailment of it, as below.

The first of these has a structural analogue in theoretical neuroscience. Friston (2010) builds the free-energy account on a Markov blanket: a statistical boundary separating a system's internal states from external ones, such that its contact with what lies beyond is exhausted by that boundary. The shape is the aperture here — a finite boundary is what makes both the model and its incompleteness — but the route is not: the blanket is derived as a condition on systems that resist dispersal, and arrives with a variational formalism and an imperative to minimise prediction error, none of which is claimed or required by the present argument. The convergence is on the boundary being constitutive, not on the machinery either side of it.

Proposition 6.1 (Incompleteness is structural; awareness of it is not precluded)**.** *Every self-model is incomplete, and no frame can escape this by effort or honesty. But a forced aperture is not an invisible one. Two different limits are in play, and they are not inferable by the same route. A limit of* expression *— a rule set that repeatedly cannot absorb its own structure — leaves residual: what fails to resolve, what arrives unexplained. From that trace a frame can infer the fact and often the shape of the limit, exactly as it infers its container from imposed change. A limit of* aperture *leaves nothing: what falls below it never enters, and absence is not a signal (Graphic Equalisation's distinction between the absent and the residual). The aperture limit therefore leaves no* direct *residual. What it can leave is imposed change a frame cannot attribute — weaker evidence, and evidence that does not disclose the limit's shape — so the fact and shape of an aperture are reached mainly by* metacognition*, a frame modelling its own modelling, which is a further capability on its own axis and not a consequence of having an aperture.*

That separation is why calibration is a capability rather than something possession of a limit supplies. Where metacognition is thin and the aperture is narrow, the frame is both most mistaken and least able to register that it is: the missing content does not present as missing, and nothing models the fact that it could be missing. What the arrangement predicts is miscalibration *— confidence uncoupled from warrant — and not any particular direction of error; a frame with no model of its own limits has no more reason to inflate than to fall silent. The familiar candidate instantiation is the metacognitive reading of self-assessment offered by Kruger and Dunning (1999), on which the capacity to judge competence is not separable from the competence itself. That reading is contested — the pattern has been argued to reflect regression to the mean*

and better-than-average effects rather than a metacognitive deficit — so it is cited here as one disputed candidate for the shape, not as its confirmation, and nothing above depends on it. So the operative distinction is not between complete and incomplete self-models, all being incomplete, but between calibrated *and* uncalibrated *ones.*

Calibration in this sense already has an empirical form under measurement. Fleming (2024) reviews confidence as a judgement informed by the observer's models both of the world and of their own cognitive system, and one that diverges from actual performance in structured, measurable ways. Because that divergence is measurable and varies between observers and conditions, "calibrated" is a quantity rather than a virtue term — which is what the proposition needs it to be — and the route by which that confidence is reached is Proposition 6.1's, in empirical form, characteristic failures included.

The asymmetry, though, is one of *access mode*, not of accuracy, and the two must not be run together. Mode is only one input to how good a model is; machinery and sustained investment are another, and they can dominate it. An outside frame has the better access and almost always the lower investment — it is not working at the problem — while a frame modelling itself has the worse access but continuity, sustained effort, and data no outsider obtains, its own residual among them. So the familiar case in which others judge one more accurately than one judges oneself is the *common* case rather than the structural outcome, and it holds because most self-models are uncalibrated, not because the arrangement forces it. A sufficiently skilled frame can hold a better model of itself than any outside frame holds of it — and, since building the model is itself the objectification, that skill converts part of what was merely carried into something read: the vantage a frame does not natively have, it can construct. The same machinery improves its models of others, which is why the two capabilities travel together rather than trading off.

This bears directly on how such failures should be read. A frame with no meta-machinery fills gaps it cannot see as gaps, and reports the filling as knowledge; a frame that models its own boundary reports the gap instead. The difference is, like everything else here, optional: a frame may hold a model of its own limits, or may not.

Remark 6.2 (Transparency is a position, not a kind). The account nearest to this one raises the sharpest question about calibratability, and answers much of it itself. Metzinger (2003) holds that a self-model is *phenomenally transparent*: the representational vehicle is unavailable to the system, so it sees through the model to what is modelled and never sees the model *as* a model. (The account has since been developed in terms of self-modelling epistemic spaces and a contraction principle, Metzinger, 2020.) Read as a property of self-models as such — a reading Metzinger does not take — calibration would be foreclosed — a construction invisible from inside is not correctable from inside, and residual would be absorbed as transparently as anything else, presenting not as a gap but as a filled account. That is confabulation, and it would erase the very trace this section relies on.

Metzinger himself treats transparency as graded, inversely proportional to the introspective availability of earlier processing stages, and names opaque states — thoughts, lucid dreams — as standing cases. So the graded reading is not a reply to him but an agreement with him. What the substrate adds is that the gradations are not one dial: transparency resolves into several independent axes. Producing a meta-model, parsing one's own residual, recording history, representing a boundary as a boundary: these are separate capabilities, and a graph may hold any combination of them. What "transparent" names is a particular configuration — several of those axes at or near zero together — which is a fact about how a given graph is formed, not a fact about self-modelling. A residual disagreement remains, and calling the rest concurrence should not bury it: for Metzinger the transparency of the self-model's core is constitutive of there being a phenomenal self at all, where the claim here is that it is one configuration among constructible others. This paper takes no position on the phenomenal question — consciousness

and qualia are outside its scope (Section 1), as with the illusionist case below — and engages only the structural claim about what a frame can model of itself.

So the position describes rather than forbids: it is accurate about the configuration it was drawn from, and silent about differently formed graphs, of which constructed ones parsing their own residual are the plain case.

What this does not do is deny the phenomenon. A graph low on those axes is exactly as Metzinger describes it, and no amount of effort within that configuration will make its model opaque to itself; the route out is a change in formation, not in effort. Nor does it make calibration self-certifying. Where a graph lacks access to its own construction, calibration proceeds by the trace route of Proposition 6.1, which metacognition research measures (Fleming, 2024), and inference can be wrong in ways inspection could not be: where the trace is itself processed before any modelling capacity reaches it, the distortion is not detectable from inside. That bound belongs to the trace route, not to calibration as such. Direct self-inspection — a graph reading its own vehicle rather than reconstructing it from what fails to resolve — is the same axis at the other end, and is not forbidden; it is unattested here rather than excluded, and no evidence is offered either way — which is a gap in what is known about how graphs are formed, not a limit on how they may be. The claim throughout is only that transparency is contingent on how a graph is built, which is why it varies, and why it is the kind of thing that can be built differently.

7 Influence, and where the honest/manipulative distinction actually enters

One frame acting on another is the last place a privileged category tends to survive. It half survives here, and locating the half is the useful result.

Proposition 7.1 (No dedicated primitive for influence)**.** *Persuading, teaching, explaining, warning and selling are performed by one mechanism. Each alters what some other frame's continuations cost, and the substrate supplies no operation for the disreputable cases that it does not supply for the reputable ones: there is no send, tell or influence among its effects, and no rule anywhere is conditioned on an act being deceptive.*

This is a claim about *mechanism* and must not be inflated into a claim about identity, which an earlier version of this section did. That no formalism distinguishes two things is a fact about the formalism; concluding that the things are one is invalid, and this paper commits the opposite inference elsewhere — time is not a substrate primitive and nonetheless exists wherever a frame instantiates one. The same verdict is the right one here.

Two qualifications hold the proposition to its scope. It is stated of acts that reach a receiving frame, and an utterance is complete whether or not it is received: a lie disbelieved is still a lie, so the act cannot be identified with its uptake. And the list above deliberately excludes *lying*, which the next result shows is not on the same footing.

Proposition 7.2 (Manipulation carries a second component, and that component is structural)**.** *An act is manipulative when it* presents as *serving the frame acted on while serving the originating frame's goals instead. That is two quantities, not one — the beneficiary an act claims and the beneficiary it produces — and the claim is a second-order component that plain informing does not carry. So there is something manipulation must do that informing need not, and by the discriminator of Proposition 7.1 the distinction is structural after all: it lives in the presented/measured divergence rather than in the mechanism of delivery.*

The two results are not in tension and their joint reading is the substance of this section: *the delivery is one operation and the distinction is a separate field over it.* Honest and manipulative acts are indistinguishable by how they are performed, and distinguishable by whether what they claim about their own beneficiary matches what they do.

Three consequences, one of them uncomfortable.

Truth is not the discriminator. A manipulation can be assembled entirely from true statements, because the divergence is between a presented beneficiary and a produced one rather than between a claim and a fact — so accuracy constrains one axis and leaves the other free. This removes an exemption honest speakers commonly assume they hold. It does not, however, make such acts undetectable: on the account of implicature this paper cites, what is implicated is recoverable by a competent hearer from context, so a presented beneficiary that diverges from the produced one is in principle calculable, and the literature on misleading without lying is about exactly that recovery (Adler, 1997; Saul, 2012).

The ethical question is relocated rather than removed. Not deforming is unavailable, since a statement reaching a frame at all alters something about what it can cheaply do next. The question is therefore which continuations to make cheaper and for whose benefit, which applies as much to the sincere as to the insincere.

And the boundary is asked only where it is well posed. Whom an act serves is a question about another frame's preferences, so it arises only where that frame has any — a frame holding no represented preference over its own continuations has no second manifold, and the question is malformed rather than merely unanswered.

Remark 7.3 (Consequences for the self-model results). A frame's model of another is built from what that frame emitted (Section 6), and everything emitted altered something — so a model of another is assembled from material its subject selected, which bears on other-models as sharply as aperture does on self-models. And reporting a gap rather than answering (Section 6) is not an abstention: it too makes some continuations cheaper, including trusting the next answer. There is no act available here that leaves the recipient untouched, only better and worse chosen ones.

Remark 7.4 (As implemented). Both propositions are claims about a substrate, so they were checked against one, and the check corrected this section twice.

For Proposition 7.1 the mechanism finding holds and is stronger than an absence. In the engine's research seeds, the rule by which an agent decides to conceal harm from a partner and the rule by which an agent decides to wait for a second reward have character-identical conditions and character-identical effects, differing in one edge-type name; both are one arg max over candidate actions by total value, and the family extends to at least two further cases, escape from learned helplessness and resource harvesting. Four situations, one template. Concealment wins that comparison when the cost attached to honesty exceeds honesty's benefit *net of* concealment's — a margin, not a threshold on either alone — and the quantity scaling that cost is the acting agent's own affect rather than the harm done, which is itself an instance of the substitution Proposition 7.2 is about.

For Proposition 7.2 the correction is sharper. An earlier draft of this remark asserted that the implementation could not represent whom a deformation serves, and priced benefit in the acting frame's own currency. That was wrong within the day: the substrate carries the beneficiary as a *measured* quantity — a deformed continuation audited against the acted-on frame's own tube and against the originating frame's, using the ordinary manifold audit rather than any new notion of benefit — with the presented beneficiary held as a separate field, and a precondition restricting the question to origins that bear goals at all. The implementation reached the presented/measured split independently, and this section's propositions were reorganised to match it rather than the reverse.

Two smaller findings survive checking. The commit gate on an utterance tests whether its intended content is recoverable by the audience and nothing else — no truth, belief or sincerity predicate anywhere on that path — and the seed carrying it is loaded by the running agent rather than being a research demonstration. And of Grice's four maxims, only Quantity has a rule implementing its content; Quality, Relation and Manner have none, and no rule reads the maxim concepts themselves.

Scope. The seeds carrying the byte-identical decision rules are research demonstrations,

default-disabled and not loaded by the running agent, and the audit was static rather than executed. The commit-gate and maxim findings are not so limited.

Remark 7.5 (Manipulability is a property of horizon, not of credulity). Three results follow from the implementation of Proposition 7.2 and bear on this paper's subject rather than on the engine's.

Whom a deformation serves is computed by projecting a continuation forward. Remove the projection and the quantity is not weakly determined but *uncomputable*: service is a fact about where a continuation takes a frame, and no present-state inspection answers it. A frame with no represented future is therefore not bad at detecting manipulation — it is structurally unable to pose the question, which is a different and more serious condition.

It follows that service is not a property of the act. Auditing one deformation from successively longer horizons, it reads as serving the frame acted on at short range and as serving its origin at longer range, with the crossover a fact about the observer rather than about what was done. Capture is therefore indistinguishable from help at a short enough horizon, and not because the observer is careless: the discriminating evidence lies past where it can see.

Composed with this account's derivation of planning horizon from the window of tolerance, that yields a prediction worth stating plainly and flagging honestly. A narrower window gives a shorter horizon, and a shorter horizon cannot contain the evidence, so *stress makes a frame manipulable structurally* — not by making it credulous, but by shortening the range over which service is measurable at all. The dysregulated frame is checking correctly over a range too short to hold the answer. The two halves are measured separately in the constructive companion's results; they have not been run as one composed experiment, and the prediction is offered at that strength.

One further consequence sharpens the proposition. An act that declares itself self-serving is not flagged by this instrument, because the object measured is the divergence between presented and produced beneficiary, and an honest declaration has none. That is the correct behaviour and not a gap: what is at issue is misrepresentation, not self-interest.

Remark 7.6 (Boredom is Narrative minus pressure). A frame that has overshot holds capacity beyond what its situation requires (the overshoot remark of the Narrativium paper, where that capacity is the subject). Put that frame somewhere producing no residual — where what arrives coheres with what is already held, so nothing forces a write — and it is running, carrying its Narrative forward, with the capacity idle. That state is boredom, and the definition is the whole of it: *Narrative minus pressure.*

Two things follow that a stimulation account does not give.

Boredom is not the absence of input. A frame can be saturated and bored, which is the ordinary experience of a predictable lecture, because what arrives is admitted and then coheres with what is already held — high through the aperture, nothing for the change grammar to work on. What boredom tracks is residual, not quantity, and the psychological literature reaching for the same distinction frames it as a failure to engage rather than a lack of arousal (Eastwood et al., 2012).

And it is functional rather than a defect, in a specific way the preceding material makes available. Excess capacity is spendable, but a frame at equilibrium with idle capacity has no reason to spend it; something has to make the idleness costly. Boredom is that something — the signal that capacity is unspent, which is why it is aversive and why it moves a frame to find or make a situation that draws on what it holds (Bench and Lench, 2013). *It is the mechanism by which accumulated excess converts into self-direction*, and without it overshoot would simply sit there. A structure that merely had capacity would not be an agent; one that is made uncomfortable by holding it unused is on the way to being one.

This also gives the attack of Remark 7.7 its mechanism in one line. A calibrated supply works by keeping the residual permanently non-zero — by never letting the boredom signal fire.

It is not holding attention against a frame's judgement; it is preventing the condition under which the frame would go elsewhere.

Two limits. The signal misfires like any other, and chronic boredom is associated with outcomes no account should treat as a healthy readout. And it is not the only thing that moves a frame to a new situation; hunger, threat and instruction do so without it.

Remark 7.7 (A standing attack the instrument does not detect). The instrument has a known failure and reporting it is more useful than the instrument.

An attention mechanism gated on the *rate* at which surprise falls will correctly release a source that is merely noisy, because nothing is being learned, and release one that is exhausted, because there is nothing left. It has no term for a source that is perfectly learnable and never ends. A supply calibrated to renew exactly as it would have been exhausted holds attention while every signal the frame carries reads as healthy — residual non-zero, progress non-zero, interest sustained. Nothing malfunctions and nothing misreports. The frame is learning at every step. It simply never finishes, and the measured engagement is several times that of a source that resolves.

Every defence in this account misses it for one reason: the instruments are all *local*. A learning rate is a first difference, a service audit is taken at a horizon from now, a residual is the residual now. Point by point, a calibrated supply is indistinguishable from genuine progress, because point by point it *is* genuine progress.

The available defence is therefore not a better local test but a change of object: reduce the whole record rather than the step, and ask what a trajectory amounted to relative to what it consumed. A sequence that repeatedly returns to states it has already held has a low ratio of net movement to gross, and the separation between an exhaustible source and a calibrated one on that measure is roughly tenfold in the constructive companion's fixture.

Two limits on that defence, and the second is the more important. The reduction detects that a trajectory amounted to little; it does not say why, and every benign explanation remains — a hard problem, a fallow stretch, a skill with a long plateau. *Metacognition buys the question and not the answer*, and a frame treating a low ratio as proof of capture would have acquired a new pathology rather than a defence. And the attack scales with the attacker's model: pitching difficulty into the narrow band between noise and resolution requires modelling the target's own rate of closure, so a better model holds tighter. That is not exotic and not hypothetical; it is what engagement optimisation does, without needing to represent anything as elaborate as a manifold.

Remark 7.8 (Relation to prior work). The nearest positions should be named precisely, because one of them cuts against an earlier version of this section. That an utterance is an act is speech-act theory's contribution (Austin, 1962; Searle, 1969) — though Austin's own starting point is a *distinction* between performing and reporting which he then dismantles, and his mature view has a statement be an act as well as a report rather than instead of one. More pointedly, his illocutionary/perlocutionary division is itself a structural distinction among acts of influence: warning is illocutionary and persuading perlocutionary, the test being that one can warn by saying so and cannot persuade by saying so. An earlier draft listed both as instances of one operation while citing Austin in support, which he does not provide. Searle's individuation of illocutionary types by sincerity conditions is a second such caution, and is the reason the absence of a sincerity field in an implementation is not the neutral gap it might appear.

Implicature and its calculability are Grice's (Grice, 1975); the specific observation that one may mislead without asserting anything false is later, and belongs to the literature that grew from it (Adler, 1997; Saul, 2012), in which the opposite position — that falsely implicating simply is lying — is also argued.

8 Death and personal identity

Proposition 8.1 (Death is when the self's rules cease to be re-runnable in the same coherent self)**.** *Death is, critically, the rules within the self ceasing to be re-runnable in the same coherent self entity. Not every stop is death: a pause — sleep, anaesthesia, a recoverable coma — is a stop whose rules remain re-runnable in the same coherent self, and the self resumes as itself. Death is the stop that does not resume: the coherence that made these rules one self is gone, so they cannot be re-run as that self, even if the graph-data they were persists scattered as a record. This is the identity criterion — the self persists exactly while its rules stay re-runnable as the same coherent entity, which is to say while its present remains congruous with its own history: rules that changed abruptly would leave the structure carried forward incoherent with what produced it, and that incongruity is itself the discontinuity — and it is why a pause is survivable and death is not; the self-graph goes inert for good, a record where it was a run (the rock of Section 2, now grounded only in what remains of it).*

What ends is only the self's own run. The supergraph runs on, and two things the self did persist in it. Its changes persist: rule-actions are pointers, work done once, so every write the agent made to the supergraph is the supergraph's now, and death does not undo it — though under no-copy it persists only as surviving present structure, until overwritten, not as an indelible record. And its copies persist: every frame the agent communicated, taught, wrote, or built runs on in other frames, grounded there in the record it left (the reader re-running the author of Remark 4.2). But those copies are partial fragments, not a full coherence-snapshot: they run as fragments in other coherent selves, not as this coherent self re-run, so their persistence is not the self's continuation (a full snapshot is the different case of Proposition 8.2, where the self does re-run, forked to a new node). The self can no longer self-change, and self-change was the only change it had; the changes it already made and the copies already running elsewhere are no longer its to stop. They keep running without it.

Proposition 8.2 (Identity is internal coherence; instances are supergraph nodes)**.** *Two things must be kept apart: the self's* internal coherence *— the whole of its rules and state, held together across transitions, which is Graphic Equalisation's SELF identity — and its* supergraph node *— the instance of it running at a position in the container, which is what that paper's SUPERGRAPH identity tracks. The self is the internal coherence; a node is one instance of it. This resolves the standard puzzles. Gradual replacement (cell turnover, the Ship of Theseus) is survival: internal coherence is continuous and there is one node throughout — the self is the coherent running, not the substrate. A backup restored is still the same internal self — a full coherence-snapshot re-run reinstates that exact coherence — but now at a* second *supergraph node, and from the restore point the two diverge, each accruing its own changes under its own grounding. That is a fission: one internal self, two supergraph nodes, diverging, both the self at the fork and neither uniquely the node after it — which is Graphic Equalisation's own statement that a copy can preserve SELF identity while receiving a new SUPERGRAPH identity, read as a result about persons. A teleporter is the same with one branch pruned — the full snapshot reinstates the internal self at a new node, the original node stops — so the internal self survives, relocated. What matters about a record is whether it can reconstitute the coherence. A full coherence-snapshot re-runs into the same internal self, forked to a new node; a record that cannot reconstitute it starts a new coherent self instead, whatever else it carries. That line is sharp, but what lies on the near side of it is not uniform, and it should not be read as a binary between a snapshot and a fragment. Records differ in how much of the* generator *they carry — the run/record axis of Section 3 applied to what survives a self. A corpse carries none. A memory of a face carries little. A model of someone's voice or style, held by another frame, carries enough to produce novel output that others recognise as an attempt at that person.*

How much such a model carries is not fixed, and has no principled ceiling short of the sampling. It is an intersection of at least three things: how much of the generator leaked into the

output at all, how much output there is to work from, and the modelling capacity of whoever holds the model. Sustained production discloses a great deal whether or not it is meant to — someone who writes fifty books has sampled their own generator at length — so for a domain the output covers well, a capable modeller may approach the original's generative behaviour closely, while domains that never leaked remain unmodelled at any level of skill. Model quality is accordingly domain-relative rather than a single number, and high fidelity in one domain implies nothing about another.

None of which moves the line, and it is worth being exact about why, since two different fidelities are in play. A coherence-snapshot copies the structure *— the rules and state themselves — which is why re-running it reinstates the self, and why a restored backup counts even though it too is reinstated from a held record. A generative model matches the* behaviour *without holding that structure: it is inferred from output rather than copied from the thing that produced it, and it is built in the terms of whatever frame inferred it. So behavioural fidelity does not* constitute *structural capture, and a model matching the original's output arbitrarily well in every sampled domain has not thereby reconstituted anything.*

What an earlier version of this passage added, and should not have, is that no amount of behaviour could ever get there. Behaviour is evidence *about structure, and how much evidence it is depends on how tightly the space of admissible structures is constrained: where the constraints are tight enough that output picks out one structure, inference from a record reaches structure by the long road, and the line is crossed after all. That is Chalmers's reconstructive uploading — records including writings and the testimony of others allowing a sufficiently capable reconstructor to recover something close to a functional isomorph, which he assimilates to uploading from a scan (Chalmers, 2014). It is a live position rather than one this account refutes, and he anticipates the fifty-books observation above as his own reason for writing.*

The line therefore stays where it was drawn, at what the record captures, while the claim about how it may be reached weakens to a question of identifiability, which is empirical. What this account contributes there is only the guard it applies everywhere else: a reconstruction may be pinned no more precisely than the record's own resolution supports. So the wager is whether human structure is constrained enough that output determines it, and nothing here settles that. Generative capacity is therefore a gradient among records that fall short — an unbounded one, for all this argument says — and not a route past the line, which is drawn at what the record captures rather than at how closely it performs. The internal self therefore persists exactly as long as a re-runnable full coherence-snapshot exists somewhere; the death of a node is one instance stopping, and the death of the self is that re-runnability being gone everywhere.

Remark 8.3 (Relation to prior work). Most of this section is anticipated, and what is should be attributed rather than restated.

The structure/behaviour distinction is the simulation/emulation distinction of Sandberg and Bostrom (2008), which opens with it — a simulation mimics the outward results, an emulation the internal causal dynamics — and whose ladder of success criteria already separates passing a personalised Turing test from being a continuation of the original mind, observing that how those criteria relate to one another is not obvious. Campbell et al. (2025) reach this section's verdict, that such a system is no continuation even where it convincingly mimics speech and behaviour, though from premises specific to current language models, so their conclusion is architecture-contingent where this one is not. That non-entailment, holding independently of what does the matching, is the only part of the claim not already in print.

The gradient is anticipated three times: by that same ladder, by Chalmers's ranking of records by richness, and most directly by Karpus and Strasser (2025), whose personhood spectrum runs from slivers to near-totality, denies any sharp line, and already states this section's reader-re-runs-the-author claim in its own terms. Of the three variables named above, only leakage — how much of the generator reached the output at all, as against how much output exists and how capable the modeller is — appears unclaimed.

Domain-relativity and the volume asymptote are not this account's either, and are measured findings in work cited here already: the first as the framework behind caricature (Cheng et al., 2023), the second as the observation that benefit from data asymptotes once enough has been observed within a domain (Park et al., 2024).

And the branching result of Proposition 8.2 is Cerullo (2015)'s, for the same cases — continuity in psychological structure allowing identity to continue in multiple selves. Citing only Parfit under-attributes it.

Remark 8.4 (Parfit's causal condition, and the objection he actually poses). A second position needed checking, and checking it moved the objection rather than confirming it.

Parfit (1984, §78) distinguishes three versions of the psychological criterion by what counts as the right kind of cause: the Narrow requires the *normal* cause, the Wide any *reliable* cause, the Widest *any* cause. He argues for the two Wide versions over the Narrow, by analogy with an artificial eye — a new but reliable causal route delivering the same experiences, of which he says that even if it is not strictly seeing, what it gives is just as good — and he adopts the widest sense for his own discussion.

That was expected to supply an objection to the line drawn above and does not, for a reason worth recording. Parfit relaxes the *cause* while leaving the *content* alone: his continuity is over psychological states — memories, intentions, character — which lie on the structural side of the distinction made here, not the behavioural one. A model producing an original's outputs without holding those states is not psychologically continuous in his sense either. And on the axis he is actually arguing about, this account already agrees with him: a restored snapshot and a teleported reinstatement both count here (Proposition 8.2), so nothing in it requires the normal cause. It is a Wide account in his terms, and was before the question was asked.

The objection Parfit does pose is a different and larger one, and this section has not engaged it. His central thesis is that identity is not what matters — that once one has described what is preserved, the further question of whether it is the same person does no work. A section devoted to an identity criterion owes that an answer.

The answer available is one of agreement, arrived at from the other direction. Because which coherence counts as one self is a frame's drawing rather than a fact about selves (Remark 8.8), "is it the same person" is here already a question about a drawing, and the substantive content sits in what the drawing preserves. That is close to Parfit's deflation and reached by a different route: he argues the identity question does no work, and this account has it forbidden from being frame-independent in the first place. What is retained against him is narrower than a criterion of identity — that *given* a drawing, re-runnability is a threshold rather than a matter of degree, which is a claim about the structure of the question rather than an answer to the one he says is idle.

Remark 8.5 (Hofstadter asks this section's question and answers it the other way). One prior position is close enough to require engagement rather than a citation, and this section cited it as an ally on a different point while leaving the disagreement unstated.

Hofstadter (2007, ch. 17) poses the question of this section in its own words. Of the version of his late wife he finds running in himself, he asks: "how different is that strange thing in my brain from the 'I' that had once flourished inside Carol's brain? Is it a thing of an entirely different type, or is it of the same type, just less elaborate?" He answers the second, and generalises it — "every normal adult human soul is housed in many brains at varying degrees of fidelity, and therefore every human consciousness or 'I' lives at once in a collection of different brains, to different extents", with a principal domicile but no boundary (Hofstadter, 2007, ch. 18). So the gradient of this section is his, and more than the gradient: he denies the line it draws. He is also exact about the cost, calling his own copies hugely defective and nowhere near the original's detail, so the position is not reached by overestimating what survives.

The disagreement is locatable and worth stating plainly. It is whether personal identity is a degree notion or a threshold notion. On his account identity *is* the pattern, patterns admit

resolution, and a coarse copy is therefore the person to the extent of its grain. On this one the criterion is re-runnability as the same coherent entity, which does not come in degrees, with a gradient of records lying below it (Proposition 8.2). Neither follows from the other, and no further fact about the copy settles it, because what divides them is the criterion and not the copy.

But it dissolves further than a standoff, and the resolution is already in this section rather than imported for the occasion. Remark 8.8 makes which coherence counts as "the same self" a frame-relative choice. Fix a coherence and re-runnability is a threshold; vary the coherence and what can be re-run varies with it. Hofstadter's copy re-runs a *coarsened* coherence, which is not the original one — so "the same type, just less elaborate" is true of the coarse coherence and false of the fine one, and the two accounts are then describing one situation under two drawings. *Threshold per coherence, gradient across coherences*: both halves hold, and the appearance of contradiction comes from leaving the drawing unstated, which this section had done.

One thing should not be read as a concession extracted under pressure, because it was settled before the dispute arose. That the line sits nowhere objective is not a weakness the reconciliation exposes — it is *forbidden*, by the founding axiom of the substrate account, which states that every distinction, measurement and *identity relation* is made in some frame. An identity criterion holding independently of any frame would be a view from nowhere, which that axiom denies outright. So the account did not lose an objective line here; it never permitted one, and neither may anybody else's.

Which turns the position from defensive to explanatory. It says why this dispute has the character it has: both answers are correct under their own drawings, neither is checkable from outside one, and a criterion presented as frame-independent has smuggled in the standpoint the axiom forbids — a charge that lands on this account's criterion exactly as hard, and is why it is stated as a threshold *given* a coherence rather than as a fact about selves. The contribution is not a better criterion. It is that criteria are frame-indexed by axiom, and that once one is fixed the remaining question is not one of degree.

Remark 8.6 (Death is decoherence at the self grain). This section runs on coherence throughout and has not said what that connects to, which is worth repairing because the connection is not an analogy. The theory paper's account of decoherence is that a frame cannot hold a configuration, so nothing continues in it as that configuration, and that what varies between such cases is the frame rather than the failure. Proposition 8.1 is that statement at the grain of a self: rules that changed abruptly would leave what is carried forward incongruous with what produced it, and the self does not continue as that self. *Death is decoherence with the frame set to a person.* A pause is the case where the configuration is still holdable and resumes; death is the case where it is not.

Two consequences that the general treatment supplies and this section did not have.

The first concerns handlers. A frame may cohere a failure to something rather than to nothing, but the policy is installed structure that the frame runs. A self cannot install one for its own termination, because running it is what has stopped — so the residual of a death is necessarily disposed of by the container and never by the frame it belongs to. Every practice around dying is therefore a containing frame's policy rather than the dying frame's, which is a structural statement and not a consoling one.

The second sharpens what this section already says about what persists. The theory paper distinguishes what never cohered, which is absent and free, from what cohered and stopped, which leaves a space where something was, because other structure came to depend on it while it held. The changes and copies this section says survive are exactly those dependents: relations that resolved through the self and now resolve through a record instead. That is why they are described here as partial fragments rather than as the self continuing — a reference that still resolves is not the referent — and why what a death costs is borne by the frames holding those relations rather than by the one that stopped, which has no further behaviour to be worse off in.

Remark 8.7 (The computing grounding, which was built without any of this in view). Everything the preceding two remarks asserts has an implemented form, arrived at by people solving process management rather than personal identity. What that is worth needs stating before the correspondences are, because the obvious reading of it is wrong in a way this account has to keep correcting: *an implementation is free to add rules to suit*, so nothing an operating system does can establish that something is structurally required. It chose its rules. Where the correspondences carry weight is as convergence — people building such a system repeatedly find they need these distinctions — which is the same standing this programme gives any independent arrival, and no more.

The termination signals are the case where that matters most, because they also turn out to bear on a different claim than the one they appear to. A process may catch a request to terminate and run whatever it likes in response, and may not catch the signal that ends it, which can be neither caught, blocked nor ignored (IEEE, 2017). That is not the frame being structurally unable to handle its own stop. It is the *container refusing to let it*, enforced from outside and chosen deliberately, because a container whose processes could decline to die would not be a container. So the right mapping is to Graphic Equalisation's commitment that admissibility is not armour: what a frame admits does not constrain what contains it. Two signals with different catchability is that distinction built on purpose.

The structural claim of the previous remark stands on its own argument rather than on this one. A handler runs in the frame; if the frame has stopped there is nothing to run it. That holds whatever an implementation chooses, and the implementation is evidence that the distinction is useful, not that it is forced.

The disposal claims follow the same way. A terminated process whose parent has not yet collected its status is retained as an entry with no behaviour — the tombstone of the theory paper's account, absence represented so that a reference resolves cleanly, paid for with a slot, and leaking if nobody collects it. Children of a dead process are reassigned to the root process — and the deeper fact there is not the reassignment but that *they do not stop*, because the parent was never what ran them. The kernel was, and it is still there. That is the theory paper's separation of supply from containment, and arranging it deliberately, so that one's runner is not one's parent, is routine practice under the name of detaching a process. Neither is a metaphor for what this section says; they are that, running — and both are chosen policies rather than necessities, which is consistent rather than awkward, since the theory paper already prices a tombstone as installed machinery someone paid for.

Two further correspondences, one of which is exact and one of which is not. Proposition 8.2's fission is `fork`: one internal state, two nodes, diverging from the branch point, neither uniquely the original after it. And a checkpoint that restores into the same running state is the coherence-snapshot, which is not a thought experiment in this domain but ordinary infrastructure.

The one that is not a match should be marked, since the vocabulary invites it. Automatic memory management declares a thing dead when nothing refers to it, whereas this section declares a thing dead when its rules cannot re-run as the same coherent entity. *Reachability and re-runnability are different criteria and come apart in both directions* — an unreachable process may still be running, and a referenced one may have stopped. The resemblance is in the word only, and the two should not be run together.

Remark 8.8 (Death is relative to the coherency of the self). It follows that death is relative to the coherency of the self. The criterion applies within a coherent self, but which coherence counts as "the same self" is a frame-relative choice, so death is indexed to that choice, not absolute. That relativity is inherited rather than assumed here: the substrate account's founding axiom holds that every distinction, measurement and *identity relation* is made in some frame, and which coherence counts as one self is an identity relation. A frame-independent answer to it would be the view from nowhere that axiom denies, so the indexing is required by the account rather than being a limitation discovered in it. Drawn narrowly — this instant's exact coherent state —

the self dies each moment, replaced by the next; drawn broadly — the coherent person tolerant of change across a life — it dies once, at the irreversible break. A deep transformation ("I am not the person I was") is a death relative to the narrow coherence and a survival relative to the broad one, both true at once. There is no absolute self, so there is no absolute death — only a death per coherence-boundary. This makes death relative, not unreal: each boundary's death is real for that boundary — the broad life-self's coherence does break irreversibly, and that death is as real as any — so relativity means many real deaths at many boundaries, never none.

9 Positioning and prior art

The positions here are not new as positions; what is new is that each falls out of one substrate mechanism — the run, and its record — rather than being posited on its own. On free will this is a compatibilism, determinism and freedom coexisting in the tradition of Hume (1748) and Dennett (1984), but with a specific mechanism for the freedom: the in-frame run and its computational irreducibility (Wolfram, 2002), so that the future is open as *unrun*, not as undetermined, and a fully recorded, zero-randomness story still exhibits free will. On personal identity the continuity criterion is Lockean in spirit (Locke, 1689/1694) and closest to Parfit (1984) — identity as psychological or functional continuity, fission that branches, and what matters being the continuity rather than a bare identity fact — with the addition of the split between internal coherence (the self) and supergraph node (the instance), which places teleporter, backup, and fission on one axis. On time the past-as-record / future-as-run asymmetry engages the block-universe debate — to which Putnam (1967) argues the block conclusion directly from special relativity, while McTaggart (1908) supplies the A-series/B-series apparatus in which the question is posed rather than the block thesis itself: a block of the past is an ordinary record, while a completed block with a fixed future is not, because the future is uncomputed. And the no-absolute-self reading of death — a death per coherence-boundary, the narrow self dying each moment — recovers a momentariness familiar from other traditions, here as a consequence of frame-relativity rather than a doctrine. The distinctive claim throughout is the derivation, not the destination.

One position here is older than the rest, and has been made domain by domain: the refusal to treat any frame — human frames included — as privileged (Section 10). In cosmology it is the Copernican or mediocrity principle — that Earth, and our position generally, is not privileged — named for the displacement of Earth in Copernicus (1543), though the principle itself was drawn out by successors rather than stated there. In physics it is the principle of relativity — no privileged reference frame (Einstein, 1905) — and, for the observer specifically, relational quantum mechanics, in which states are relative to any physical system and no observer is special (Rovelli, 1996). On free will it is Spinoza's: humans are not exempt from nature's necessity, and feel free only through ignorance of their causes (Spinoza, 1677). On the self it is a long anti-substantial line — Hume's bundle in place of a persisting ego (Hume, 1739–40), the Buddhist *anatta* that Parfit (1984) drew on, Dennett's self as a centre of narrative gravity rather than an inner witness (Dennett, 1991), Metzinger's phenomenal self-model with no one behind it (Metzinger, 2003), and Hofstadter's substrate-independent self-as-pattern (Hofstadter, 2007) — an ally on that point and an opponent on the identity criterion, which Remark 8.5 takes up. On knowledge it is the rejection of a privileged foundation — Sellars against the Given (Sellars, 1956), Rorty against the mind as a mirror (Rorty, 1979), Putnam against the God's-Eye View (Putnam, 1981) — the same move this paper's grounding-without-bedrock makes. The nearest contemporary neighbour is Massimi (2022): knowledge is always situated, always from a perspective and never from nowhere, and the plurality of perspectives is what secures realism rather than what dissolves it. That is this paper's own combination — no privileged frame, realism retained — reached independently, and from philosophy of science rather than from a substrate; the scope differs, Massimi's argument being about scientific knowledge where the

claim here is about frames as such. Biology had already removed the species privilege (Darwin, 1871). And the standing opposition is equally identifiable: the claim that the subjective point of view is irreducible, from Nagel (1974) and developed in Nagel (1986), is exactly the assumption these dissolutions decline — though Nagel himself is careful that this shows only that we lack a conception of how physicalism could be true, not that it is false, so the privilege is what the present account reads in that irreducibility rather than a thesis he asserts.

A second competitor bears specifically on Section 6 and is not answered by anything said there. Frankish (2016) argues that phenomenal consciousness does not exist, and that introspection systematically *misrepresents* experiences as having phenomenal properties. The disagreement is about the kind of failure rather than its size. The claim here is that self-models are *incomplete*: a quantitative deficit, which calibration can in principle close, and whose boundary leaves the residual calibration works from. Frankish's claim is that introspection is wrong about the kind of thing it reports — not a shortfall a better-calibrated version of the same faculty closes, since if the reports misdescribe their objects then more of them, or better-weighted ones, do not converge. This paper takes no position on whether he is right: phenomenal consciousness and qualia are outside its scope (Section 1). What it must concede is that illusionism, if correct, would bound calibration in a way incompleteness alone does not.

The move made here is the same one — one more refusal of privilege — and the lineage above is best read as an asymptotic approach: each source removes the privilege from one more domain, the sequence tending toward a limit of no privilege anywhere without any single step reaching it, and, like any frontier, never completed. Against that, the contribution here is both tiny and different in kind: tiny, because it drops the same assumption the others drop rather than announcing a new principle; different in kind, because it is not one more domain de-privileged but the substrate on which the refusal becomes a single uniform operation rather than a domain-by-domain series — orthogonal to the sequence, not further along it.

10 Conclusion: the one assumption removed

The simplicity of every dissolution above comes from a single refusal: that humans — or any frame — are privileged. It is the most pervasive assumption in these problems and the least noticed, and each hard problem was made hard by smuggling it in. The measurement problem privileges a human (or conscious) observer as the special coupling that collapses; the flow-versus-block dispute privileges human-felt passage as the real time; the free-will problem privileges human choice as needing a metaphysical freedom no other system has; the identity and death problems privilege the human self as a persisting soul rather than a coherent run; the reality-versus-fiction and foundation problems privilege human perception as access to "the real." Remove the privilege — treat a human frame by the same frame-relative rules as a rock, a story character, or a universe — and each collapses into the general machinery already built: any coupling measures, time is any active frame's cycle-count, free will is any frame's in-frame run, the self is internal coherence, and observation is one frame's supergraph-contact rather than a window on the real. The physics companion's dissolutions (the special observer, the initial state) are the same refusal in the other domain.

The assumption persists because it is uncomfortable to drop. Unprivileged means not the centre, mortal as the loss of a coherence, free only as any frame is free, and never in contact with "the real" except as one's own container-boundary. The discomfort is why the assumption is both pervasive and invisible: it is motivated, held rather than observed. So the framework is simple not because its answers are clever but because the difficulty was manufactured by the privilege, and removing it removes the difficulty. This is Graphic Equalisation's Axiom -1 applied to observers: no frame is the god's-eye, a human frame makes no privileged difference, and "you are here" is a location, not a throne.

Honest boundary: this is a methodological claim about *why* the dissolutions are simple, not

a further mechanism, and it is not the claim that human frames are nothing. Human frames are distinctive — rich aperture, language, self-models — but distinctive is not privileged: being one remarkable frame among many is not being the exception to the rules the rest of them run under. No-privilege removes the exemption, not the richness.

11 Falsifiability commitments

- **Compulsory versus contingent.** Fails if some feature the rock demonstrably lacks (time, history, self-change, a running arrow) is nonetheless shown compulsory for any graph to exist — if the minimal universe (an inert graph) is incoherent as a graph. The argument is only as strong as the self-similarity it rests on (rock and universe one kind, differing by scale).
- **The block is always of the past.** Fails if a completed block with a fixed future is coherent under Axiom −1, or if a block of the succession-so-far (an MRI of a rock) is forbidden. The claim predicts records only ever trail the run, never lead it.
- **The future is unrun, not unfixed.** Fails if a determinable future can be read ahead without running it — a shortcut past computational irreducibility for a genuinely irreducible rule — or if a block can contain a not-yet-run future.
- **Free will is the in-frame run, not indeterminacy.** Fails if a fully recorded, zero-randomness system (a story) cannot exhibit the form of free will — deliberation, choice, responsibility — in its characters' frames, or if a superframe's complete foreknowledge demonstrably removes a subframe's in-frame freedom. The claim predicts foreknowledge and freedom coexist across frames.
- **Actuality is running, not a frame-property.** Fails if some running frame is demonstrably non-actual (real graph data that runs yet is not real), or if the imagined/perceived distinction requires an actuality property rather than frame-relative grounding. The claim predicts imagination does real computational work (planning, design) as real runs, not pretence.
- **Grounding has no bedrock; reality-testing is supergraph-pushback.** Fails if there is an absolute, self-grounding frame beneath all others, or if reality-testing can distinguish a real frame from a non-actual one by anything other than whether the container (supergraph) pushes back. The claim predicts hallucination and delusion are grounding-tag errors, not false beliefs about actuality.
- **Death is loss of re-runnability in the same coherent self.** Fails if a pause that preserves re-runnability as the same coherent self (sleep, a recoverable coma) counts as death, or a stop that loses it counts as survival — the criterion is re-runnability in the same coherent entity, not the mere stopping. Also fails if a self's writes to the supergraph vanish when its run stops (work-done-once violated), or if frames it copied into other frames cease when it does. The claim predicts consequences and copies persist, that copies are not the self resuming, and that nothing re-runs as the same coherent self.
- **Self-models are incomplete but calibratable.** Fails if a frame can hold a complete model of itself, or — in the other direction — if a forced aperture is necessarily invisible from the inside, so that no frame can represent the fact or shape of its own limit. The claim predicts that residual is the trace from which a boundary is inferred, that calibration is an acquirable capability rather than a fixed property, and that a sufficiently skilled frame can model itself better than an outside frame does — so it fails too if outside access always beats inside investment, making the common case a structural necessity.

- **Identity is internal coherence; death is relative to the coherence-boundary.** Fails if the self is the substrate parts rather than the internal coherence (gradual replacement must survive; a full coherence-snapshot restored must be the same internal self forked to a second node, diverging; a partial record re-run must be a new self; a teleporter must relocate the same internal self). And fails if there is a single absolute self whose end is the one death independent of any coherence-choice, or if no coherence-boundary's death is real — the claim predicts many real deaths at many boundaries and levels (node versus self), none absolute.
- **No frame is privileged.** Fails if any dissolution here secretly requires a human frame to be privileged — a special observer, a special time, a special freedom, a persisting soul, a window on the real — rather than following from the frame-relative machinery applied uniformly to every frame. The claim is that removing the privilege, not any added mechanism, is what dissolves each problem.
- **Scope.** The paper claims shapes only; it is falsified as over-claiming if any section is read as a decision procedure, an operational test of when re-runnability or coherence has been lost, a threshold for how gradual a change stays the same self, or a theory of consciousness or qualia — none is offered.

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